

Consensus-Based Formation Control with Collision Avoidance for Non-Holonomic Multi-Agent Systems

Jesús Leonardo Cárdenas Martínez
Unidad Tamaulipas
Cinvestav
Ciudad Victoria, México
jesus.cardenas@cinvestav.mx

José Gabriel Ramírez Torres*
Unidad Tamaulipas
Cinvestav
Ciudad Victoria, México
grtorres@cinvestav.mx*

Abstract—This paper presents a distributed control scheme for formation navigation of multi-agent systems composed of differential-drive mobile robots.

Unlike conventional approaches that model agents as holonomic systems, the proposed scheme explicitly preserves the non-holonomic kinematic constraints of each robot through an exponential control law. Inter-agent coordination is achieved via an error-based consensus protocol that generates reference velocities consistent with each robot's dynamics, without requiring linearization or kinematic model simplification.

Inter-agent collision avoidance is incorporated through the Feasible Velocities Polygon (FVP), an anisotropic representation constructed from LiDAR sensor measurements that maps each detected neighboring agent as a linear constraint in the robot's velocity space (v, ω) , in contrast to the isotropic potential fields commonly employed in the literature.

Three scenarios are validated in a simulation environment: average consensus with collision avoidance, leader-following, and V-formation maintenance. Results demonstrate the convergence of all agents toward the desired formation objectives while preserving collision-free trajectories in all three cases.

Index Terms—component, formatting, style, styling, insert.

I. INTRODUCTION

Multi-agent systems operating in formation have attracted significant research attention due to their potential in applications such as search and rescue, surveillance, and industrial automation [1].

Coordinating a group of mobile robots to maintain a desired geometric configuration while navigating among other agents remains, however, an open problem. Most existing approaches model agents as holonomic particles, abstracting away the kinematic constraints inherent to real robotic platforms. While this simplification yields tractable control laws, it compromises the fidelity of the resulting solutions when deployed on differential-drive robots, one of the most widely used platforms in industrial navigation tasks.

Consensus-based protocols have emerged as a principled framework for multi-agent coordination, providing convergence guarantees under mild assumptions on the communication topology. Formation control strategies, including leader-follower, behavioral, and virtual structure approaches, have

been shown to be expressible within this unified framework [2].

The communication structure of such systems is naturally described through graph-theoretic tools [3], which characterize how information flows among agents. Nevertheless, the vast majority of consensus-based formation controllers are derived for first- or second-order integrator dynamics, leaving the non-holonomic case largely unaddressed.

This work proposes an extension of an existing autonomous navigation law for differential-drive robots [4] toward a multi-agent formation control scenario. The proposed scheme leverages consensus protocols to coordinate local interactions among agents, while explicitly preserving the non-holonomic kinematic constraints of each robot. Inter-agent collision avoidance is incorporated through the Feasible Velocities Polygon (FVP), an anisotropic representation that maps each neighboring agent detected by the LiDAR sensor as a linear constraint in the robot's velocity space (v, ω) [4], in contrast to the isotropic potential fields commonly found in the literature [5].

A. Consensus-Based Formation Control

Consensus as a formal framework for the coordination of mobile agents was established through the study of convergence under nearest-neighbor interaction rules, without the need for a centralizing agent [6]. This work laid the theoretical foundations for convergence analysis in multi-agent systems under dynamic communication topologies. The continuous-time consensus protocol for agent i is expressed as:

$$\dot{x}_i = \sum_{j \in \mathcal{N}_i} (x_j - x_i) \quad (1)$$

where $x_i \in \mathbb{R}^n$ represents the state of agent i and $\mathcal{N}_i$ denotes the set of neighbors with which agent i maintains active communication. Under this protocol, all agents converge asymptotically to a common state when the communication topology is sufficiently connected, as each agent continuously

reduces its disagreement with respect to its local neighborhood. Consensus was subsequently extended to second-order dynamics, incorporating both position and velocity as consensus variables [2]. Furthermore, it has been shown to constitute a unifying framework encompassing behavior-based, virtual structure, and leader-follower approaches as particular cases [2], motivating the development of increasingly enriched protocols capable of addressing scenarios of greater complexity.

II. PROBLEM FORMULATION

The coordination of differential-drive robots in formation presents two interconnected challenges: the non-holonomic nature of each agent's kinematics, and the need to avoid inter-agent collisions during navigation. The following subsections address each of these aspects and motivate the proposed approach.

A. System Orders in Consensus: The Non-Holonomic Gap

Consensus protocols have been studied for first-order systems [7] and extended to second-order dynamics [8]. However, the vast majority of these developments assume holonomic agents whose control inputs can be directed arbitrarily in any direction of the state space.

This assumption is inadequate for differential-drive robots, whose kinematics impose non-holonomic constraints that fundamentally limit the available directions of instantaneous motion. Overcoming this difficulty requires qualitatively different control strategies [9], and although recent works have addressed extensions with input constraints [10] and second-order formulations [11], the problem remains open within the distributed consensus framework.

B. Non-Holonomic Systems and Differential-Drive Robots

A non-holonomic system presents non-integrable velocity constraints [12]. The differential-drive robot is one of the most representative examples of this class. Its configuration is defined by $\mathbf{q} = [x \ y \ \theta]^T$, and its kinematic model is:

$$\dot{\mathbf{q}} = \mathbf{J}(\mathbf{q})\mathbf{u} = \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} \quad (2)$$

where v is the linear velocity and ω the angular velocity. The non-holonomic constraint is expressed as:

$$\dot{y} - \dot{x} \tan \theta = 0 \quad (3)$$

Despite their widespread industrial use, differential-drive robots are frequently treated as point particles or artificially converted into holonomic systems through an auxiliary point [13].

An exponential control law that directly respects these constraints was developed in [4]. The error is defined in polar coordinates as $\mathbf{z} = [a \ \alpha]^T$, where:

$$a = \sqrt{x_e^2 + y_e^2}, \quad \alpha = \arctan 2(y_e, x_e) + \theta_e \quad (4)$$

The resulting control law is:

$$\begin{cases} v = k_1 a \cos \alpha \\ \omega = k_2 \alpha + k_1 \sin \alpha \cos \alpha \end{cases} \quad (5)$$

with $k_1, k_2 > 0$. Global exponential stability of the origin ($a = 0, \alpha = 0$) is established via the Lyapunov function $V(\mathbf{z}) = \frac{1}{2}a^2 + \frac{1}{2}\alpha^2$ [4], constituting the foundation of the proposed approach.

C. Collision Avoidance: Isotropic Approaches

Inter-agent collision avoidance has been addressed predominantly through artificial potential fields [5], where each agent generates an isotropic repulsive function acting on neighboring agents, treated as point particles. This strategy has been widely adopted in multi-agent formation control [14], [15]; however, its isotropic nature is fundamentally incompatible with differential-drive robots, whose response to neighboring agents depends critically on their current orientation [16]. In contrast, [4] introduces the Feasible Velocities Polygon (FVP), which maps each detected agent as a linear constraint on (v, ω) , resulting in an inherently anisotropic representation that preserves the non-holonomic nature of the system.

III. PROPOSED APPROACH

The proposed scheme operates as a three-stage pipeline, as illustrated in Fig. 1. A consensus protocol processes neighborhood state information to produce an error vector $\mathbf{z} = [a, \alpha]^T$, which feeds the exponential control law to generate a reference velocity $\mathbf{u}_{\text{ref}} = [v, \omega]^T$. This reference is then passed to the Feasible Velocities Polygon module, which incorporates LiDAR measurements to enforce collision avoidance and outputs the admissible command $\mathbf{u}^*$.

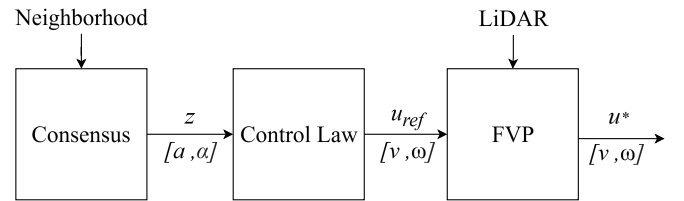


Fig. 1. Block diagram of the proposed control scheme. The consensus module processes neighborhood information and produces the error vector $\mathbf{z} = [a, \alpha]^T$, which drives the exponential control law to generate a reference velocity $\mathbf{u}_{\text{ref}}$. The FVP module incorporates LiDAR measurements to enforce collision avoidance and outputs the admissible command $\mathbf{u}^*$.

A. Graph-Theoretic Communication Models

Let $\mathcal{S} = \{1, \dots, n\}$ be the set of n differential-drive robots. The communication structure of the system is modeled as a directed graph $\mathcal{G}(t) = (\mathcal{S}, \mathcal{E}(t))$, where $\mathcal{E}(t) \subseteq \mathcal{S} \times \mathcal{S}$ is the set of active edges at time t . A pair $(j, i) \in \mathcal{E}(t)$ indicates that agent i receives information from agent j at instant t . The neighborhood of agent i is defined as:

$$\mathcal{N}_i(t) = \{j \in \mathcal{S} : (j, i) \in \mathcal{E}(t)\} \quad (6)$$

The weighted adjacency matrix $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{n \times n}$ is defined as $a_{ij} > 0$ if $(j, i) \in \mathcal{E}(t)$ and $a_{ij} = 0$ otherwise, with $a_{ii} = 0$. The associated Laplacian matrix $\mathcal{L} = [l_{ij}] \in \mathbb{R}^{n \times n}$ is defined as:

$$l_{ij} = \begin{cases} \sum_{k \neq i} a_{ik} & \text{if } i = j \\ -a_{ij} & \text{if } i \neq j \end{cases} \quad (7)$$

To guarantee convergence of the consensus protocol, $\mathcal{G}(t)$ must contain a directed spanning tree, a necessary and sufficient condition for $\mathcal{L}$ to have exactly one zero eigenvalue [2], [3]. The communication topology is updated dynamically according to:

$$(j, i) \in \mathcal{E}(t) \iff \|\mathbf{p}_i(t) - \mathbf{p}_j(t)\| \leq r_c \quad (8)$$

where $r_c > 0$ is the communication radius and $\mathbf{p}_i = [x_i \ y_i]^T$ is the position of agent i in the plane.

B. Kinematic Model

The multi-agent system is composed of n differential-drive robots, each modeled as a unicycle system. For agent $i \in \mathcal{S}$, the kinematic model is:

$$\dot{\mathbf{q}}_i = \mathbf{J}(\mathbf{q}_i) \mathbf{u}_i = \begin{bmatrix} \cos \theta_i & 0 \\ \sin \theta_i & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_i \\ \omega_i \end{bmatrix} \quad (9)$$

where $\mathbf{q}_i = [x_i \ y_i \ \theta_i]^T \in \mathbb{R}^2 \times \mathbb{S}^1$ is the configuration vector and $\mathbf{u}_i = [v_i \ \omega_i]^T$ the control input, subject to:

$$|v_i| \leq v_{\max}, \quad |\omega_i| \leq \omega_{\max} \quad (10)$$

The non-holonomic constraint of each agent is implicitly preserved in the structure of $\mathbf{J}(\mathbf{q}_i)$, which prevents instantaneous lateral displacement. The control law (5) generates trajectories that simultaneously correct position and orientation, converging exponentially toward the desired reference, as illustrated in Fig. 2.

C. Consensus Protocol and Formation Strategy

An error-based consensus protocol is proposed to provide the input variables of the control law defined in (5). Since this law operates on a distance a_i and a relative angle α_i , the consensus error of agent i is defined as the weighted average of position differences with respect to its active neighbors:

$$\mathbf{e}_i = \begin{bmatrix} e_x \\ e_y \end{bmatrix} = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} a_{ij}(t) \begin{bmatrix} x_j - x_i \\ y_j - y_i \end{bmatrix} \quad (11)$$

where $|\mathcal{N}_i|$ denotes the cardinality of the neighborhood of agent i . From this error vector, the variables required by the control law are obtained:

$$a_i = \sqrt{e_x^2 + e_y^2} \quad (12)$$

$$\alpha_i = \text{atan2}(e_y, e_x) - \theta_i \quad (13)$$

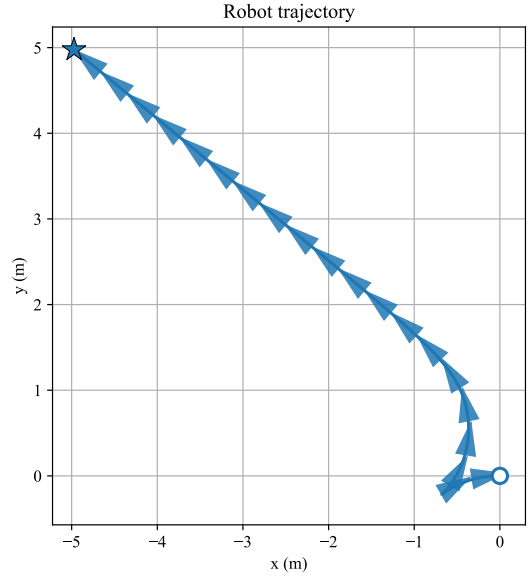


Fig. 2. Trajectory generated by the exponential control law for a differential-drive robot navigating from point A to point B. The curve reflects the simultaneous correction of position and orientation characteristic of the unicycle model under non-holonomic constraints.

The reference velocities for each follower are obtained by substituting (12) and (13) into (5):

$$\mathbf{u}_{i,\text{ref}} = \begin{bmatrix} v_{i,\text{ref}} \\ \omega_{i,\text{ref}} \end{bmatrix} = \begin{bmatrix} k_1 a_i \cos \alpha_i \\ k_2 \alpha_i + k_1 \sin \alpha_i \cos \alpha_i \end{bmatrix} \quad (14)$$

For leader-following with formation maintenance, an offset $\psi_i = [\psi_i^x \ \psi_i^y]^T \in \mathbb{R}^2$ is defined for each follower $i \in \mathcal{S}$, representing its desired position relative to the leader. Based on the consensus protocol proposed in [17], the following modification is introduced: rather than applying consensus on raw positions, the protocol operates on the offset-shifted positions $(\mathbf{x}_i - \mathbf{r}_i)$, ensuring that the desired formation is an exact equilibrium point. The state of each follower evolves as:

$$\dot{\mathbf{x}}_i = - \sum_{j \in \mathcal{N}_i} a_{ij} [(\mathbf{x}_i - \mathbf{r}_i) - (\mathbf{x}_j - \mathbf{r}_j)] - \beta b_i^0 (\mathbf{x}_i - \mathbf{x}_0 - \mathbf{r}_i) \quad (15)$$

where $b_i^0 \in \{0, 1\}$ indicates whether agent i has direct communication with the leader and $\beta > 0$ is the attraction gain. At equilibrium, $\mathbf{x}_i = \mathbf{x}_0 + \mathbf{r}_i$ for all $i \in \mathcal{S}$, and the consensus term vanishes identically. The protocol objective is to guarantee:

$$\lim_{t \rightarrow \infty} [\mathbf{p}_i(t) - \mathbf{p}_0(t) - \mathbf{r}_i] = \mathbf{0}, \quad \forall i \in \mathcal{S} \quad (16)$$

The proposed protocol induces in each robot a reference velocity $\mathbf{u}_{i,\text{ref}}$ consistent with its non-holonomic dynamics, which is subsequently filtered by FVP_i to guarantee inter-agent collision avoidance.

D. Anisotropic Collision Avoidance

For agent $i \in \mathcal{S}$, the environment is perceived through a LiDAR sensor whose frame is displaced from the robot frame by the vector $\mathbf{r}_p = [r_p \ 0]^T \in \mathbb{R}^2$, with displacement only along axis X_{r_i} .

The sensor provides a set of measurements $\mathcal{O}_i = \{(d_{ik}, \phi_{ik})\}$, where d_{ik} is the distance to the point detected by ray k and ϕ_{ik} is its angle in the robot frame. The influence zone is bounded by $d_l > 0$, so that only measurements with $0 < d_{ik} \leq d_l$ generate active constraints.

Each active measurement is mapped as a linear constraint in the velocity space (v_i, ω_i) through the velocity damper model [4]:

$$A_{ik} v_i + B_{ik} \omega_i \leq C_{ik} \quad (17)$$

where the coefficients are obtained from the rigid-body kinematics of the robot-sensor system:

$$\begin{aligned} A_{ik} &= -\cos(\phi_{ik}), \\ B_{ik} &= -\|\mathbf{r}_p\| \sin(\phi_{ik}), \\ C_{ik} &= -\xi \frac{d_{ik} - d_s}{d_l - d_s} \end{aligned} \quad (18)$$

with $d_s > 0$ the safety distance and $\xi > 0$ a convergence coefficient. The term $B_{ik} = \|\mathbf{r}_p\| \sin(\phi_{ik})$ captures the rotational contribution of ω_i evaluated at the sensor mounting point, resulting from applying rigid-body kinematics to vector $\mathbf{r}_p$ [4]. The set of constraints (17) together with the velocity limits (10) defines the Feasible Velocities Polygon of agent i :

$$\text{FVP}_i = \left\{ \mathbf{u}_i \in \mathbb{R}^2 \left| \begin{array}{l} A_{ik} v_i + B_{ik} \omega_i \geq C_{ik}, \quad \forall k \in \mathcal{O}_i \\ |v_i| \leq v_{\max}, \quad |\omega_i| \leq \omega_{\max} \end{array} \right. \right\} \quad (19)$$

The admissible control vector $\mathbf{u}_i^* = (v_i^*, \omega_i^*)$ is obtained as the projection of the reference velocity $\mathbf{u}_{i,\text{ref}}$ onto FVP_i :

$$\mathbf{u}_i^* = \arg \min_{\mathbf{u}_i \in \text{FVP}_i} \frac{1}{2} \|\mathbf{u}_i - \mathbf{u}_{i,\text{ref}}\|^2 \quad (20)$$

This formulation is inherently anisotropic, as each constraint depends on the angle ϕ_{ik} of the detected ray with respect to the robot frame, capturing the directional nature of the non-holonomic system in its interaction with neighboring agents.

IV. SIMULATION RESULTS

The proposed approach was validated in Gazebo under a ROS2 framework. A system of differential-drive robots was deployed in a planar workspace, with all agents initialized at random positions. The parameters used in each experiment are summarized in the corresponding tables.

TABLE I
PARAMETERS FOR THE AVERAGE CONSENSUS EXPERIMENT

Parameter	Symbol	Value
Linear gain	k_1	0.8
Angular gain	k_2	1.0
Maximum linear velocity	$v_{\max}$	1.0 m/s
Maximum angular velocity	$\omega_{\max}$	1.0 rad/s
Influence distance	d_l	3.0 m
Safety distance	d_s	0.85 m
Convergence coefficient	ξ	1.0
Sensor offset	$\ \mathbf{r}_p\ $	0.25 m

A. Average Consensus with Collision Avoidance

The experiment considers $N = 8$ differential-drive robots randomly distributed in a workspace of approximately 8×8 m. Each robot updates its reference velocity from the consensus error defined in (11), which quantifies the position disagreement with respect to its active neighbors. This reference is subsequently projected onto FVP_i , ensuring that the executed velocities are admissible with respect to the neighboring agents detected by the LiDAR sensor. As observed in Fig. 3, all robots converge toward the average of their initial positions in approximately 30 s, preserving collision-free trajectories throughout the process. The time evolution plots confirm the asymptotic convergence of both coordinates toward a common value, consistent with the diffusion dynamics induced by the proposed consensus protocol.

B. Leader-Following with Collision Avoidance

TABLE II
PARAMETERS FOR THE LEADER-FOLLOWING EXPERIMENT

Parameter	Symbol	Value
Linear gain	k_1	1.5
Angular gain	k_2	1.5
Leader attraction gain	k_L	1.0
Maximum linear velocity (followers)	$v_{\max}$	0.8 m/s
Maximum angular velocity (followers)	$\omega_{\max}$	0.8 rad/s
Maximum linear velocity (leader)	$v_{\max}^{(0)}$	0.25 m/s
Maximum angular velocity (leader)	$\omega_{\max}^{(0)}$	0.5 rad/s
Waypoint tolerance	ε	0.15 m

The experiment considers $N = 5$ differential-drive robots, where robot0 acts as the leader and the remaining four as followers. The leader sequentially traverses the vertices of a 5×5 m square defined by the waypoints $(5, 0) \rightarrow (5, 5) \rightarrow (0, 5) \rightarrow (0, 0)$, starting from the origin with orientation $\theta_0 = 0$. The consensus error of each follower $i \in \mathcal{S}$ incorporates both the influence of its neighbors and the attraction toward the leader position:

$$\mathbf{e}_i = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \begin{bmatrix} x_j - x_i \\ y_j - y_i \end{bmatrix} + k_L \begin{bmatrix} x_0 - x_i \\ y_0 - y_i \end{bmatrix} \quad (21)$$

where $k_L > 0$ is the attraction gain toward the leader and (x_0, y_0) is its current position. As shown in Fig. 4, the leader traverses the square trajectory while the followers are pulled toward its position, reactively adapting their trajectories to the leader's direction changes. The time evolution plots reflect the

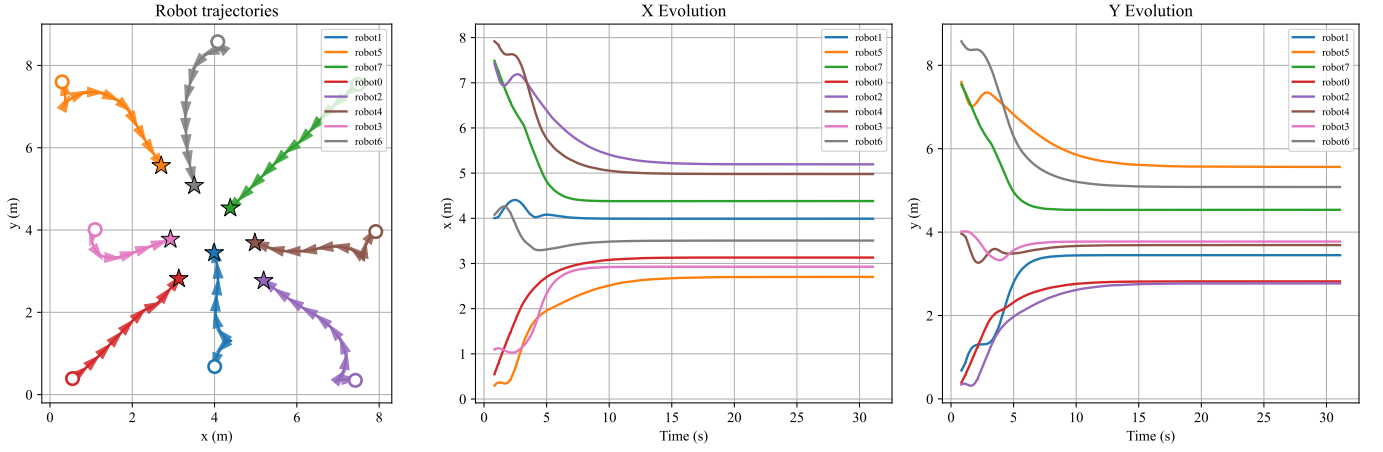


Fig. 3. Simulation results for the average consensus experiment with $N = 8$ differential-drive robots. Robot trajectories (left) and time evolution of the x and y coordinates (center and right). All agents converge toward the average of their initial positions while the FVP module ensures collision-free trajectories throughout the motion.

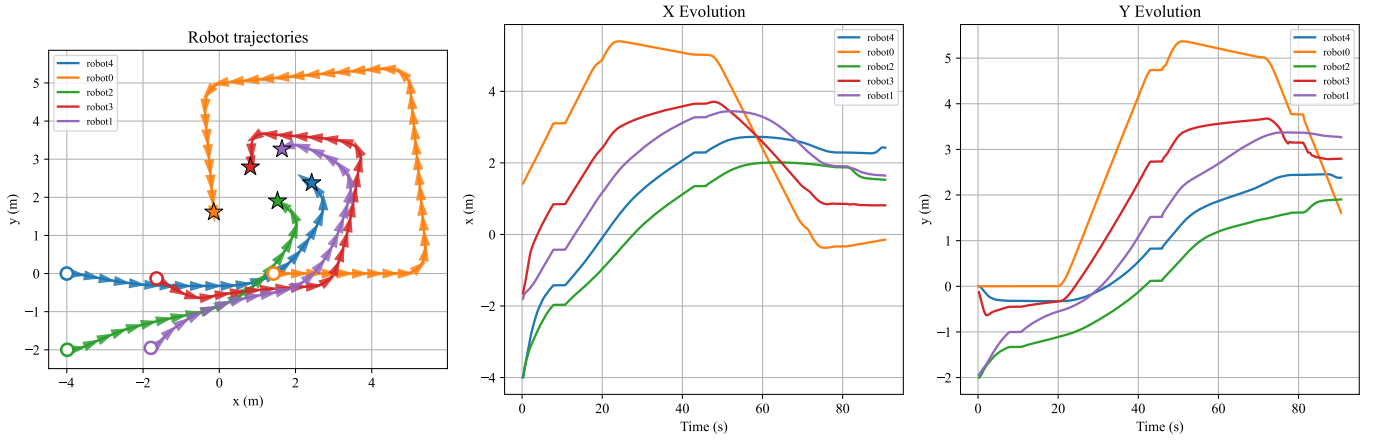


Fig. 4. Simulation results for the leader-following experiment with $N = 5$ differential-drive robots. The leader (robot0) traverses a 5×5 m square trajectory while the followers converge toward its position via the proposed consensus protocol. Stars indicate the final positions of each agent.

progressive convergence of the followers toward the leader state, with a transient of approximately 90 s associated with the length of the trajectory and the imposed velocity limits.

C. Formation Maintenance with Leader Following

TABLE III
PARAMETERS FOR THE FORMATION MAINTENANCE EXPERIMENT

Parameter	Symbol	Value
Linear gain	k_1	1.0
Angular gain	k_2	1.5
Anchor attraction gain	β	2.0
Leader orientation filter	γ	0.05
Maximum linear velocity (followers)	$v_{\max}$	0.8 m/s
Maximum angular velocity (followers)	$\omega_{\max}^{(0)}$	0.8 rad/s
Maximum linear velocity (leader)	$v_{\max}^{(0)}$	0.25 m/s
Maximum angular velocity (leader)	$\omega_{\max}^{(0)}$	0.5 rad/s
Waypoint tolerance	ε	0.15 m

The experiment considers $N = 5$ differential-drive robots under a leader-follower topology, where robot0 sequentially

TABLE IV
V-FORMATION OFFSETS AT STEADY STATE ($\theta_L = 0$)

Robot	Arm	Level	$\mathbf{r}_i$ (m)
robot1	left	1	$(-0.707, +0.707)$
robot2	left	2	$(-1.414, +1.414)$
robot3	right	1	$(-0.707, -0.707)$
robot4	right	2	$(-1.414, -1.414)$

traverses the vertices of a 5×5 m square defined by the waypoints $(5, 0) \rightarrow (5, 5) \rightarrow (0, 5) \rightarrow (0, 0)$.

The formation offsets $\mathbf{r}_i$ are computed dynamically by rotating a fixed arm vector by the leader's filtered orientation θ_L , with opening angle $\varphi_V = \pi/4$ and inter-robot spacing $d = 1.0$ m, as detailed in Table IV.

A low-pass filter with coefficient $\gamma = 0.05$ is applied to θ_L to avoid abrupt offset updates at direction changes.

Each follower updates its state via the modified protocol (15), where the consensus term operates on offset-shifted positions $(\mathbf{x}_i - \mathbf{r}_i)$, guaranteeing that the desired formation is

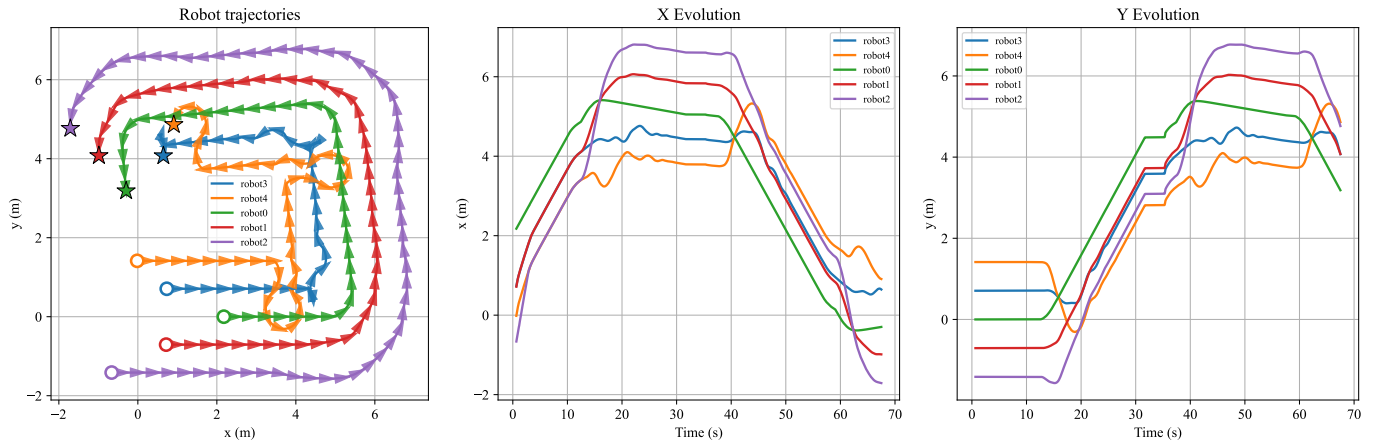


Fig. 5. Simulation results for the V-formation maintenance experiment with $N = 5$ differential-drive robots. The leader (robot0) traverses a 5×5 m square trajectory while the followers preserve the desired formation geometry via the protocol (15). Stars indicate the final positions of each agent.

an exact equilibrium. As observed in Fig. 5, the followers preserve the V-formation geometry throughout the leader's trajectory, reactively adapting to direction changes. The time evolution plots of both coordinates reflect the joint system dynamics, with a transient of approximately 70 s associated with the trajectory length and the imposed velocity limits.

V. CONCLUSION

A distributed control scheme for formation navigation of differential-drive multi-agent systems was presented, integrating an error-based consensus protocol with the Feasible Velocities Polygon for inter-agent collision avoidance. Unlike existing approaches, the proposed scheme explicitly preserves the non-holonomic kinematic constraints of each agent without resorting to linearization or model simplification. The anisotropic representation of neighboring agents through the FVP, constructed from LiDAR measurements, constitutes a fundamentally different alternative to isotropic potential fields, capturing the directional nature of the differential-drive robot in its interaction with the environment. Three simulation scenarios of increasing complexity were validated in Gazebo under a ROS2 framework: average consensus with collision avoidance, leader-following, and V-formation maintenance with dynamic offset rotation. In all cases, the agents converged toward the desired formation objectives while preserving collision-free trajectories.

Future work will extend the scheme to environments with static and dynamic obstacles, explore richer consensus protocols, and investigate the incorporation of a final orientation constraint into the control law, which would influence agent trajectories and enable orientation-aware consensus.

REFERENCES

- [1] Q. Chen, Y. Wang, Y. Jin, T. Wang, X. Nie, and T. Yan, "A survey of an intelligent multi-agent formation control," *Applied Sciences*, vol. 13, no. 10, p. 5934, 2023.
- [2] W. Ren, "Consensus based formation control strategies for multi-vehicle systems," in *Proceedings of the 2006 American Control Conference*, 2006, pp. 4237–4242.
- [3] M. Mesbahi and M. Egerstedt, *Graph Theoretic Methods in Multiagent Networks*. Princeton University Press, 2010.
- [4] G. Ramírez and S. Zeghloul, "Collision-free path planning for nonholonomic mobile robots using a new obstacle representation in the velocity space," *Robotica*, vol. 19, pp. 543–555, 2001.
- [5] O. Khatib, "Real-time obstacle avoidance for manipulators and mobile robots," *IEEE Transactions on Robotics and Automation*, vol. 5, no. 1, pp. 90–98, 1986.
- [6] A. Jadbabaie, J. Lin, and A. S. Morse, "Coordination of groups of mobile autonomous agents using nearest neighbor rules," *IEEE Transactions on Automatic Control*, vol. 48, no. 6, pp. 988–1001, 2003.
- [7] Z. Yang, X. Pan, Q. Zhang, and Z. Chen, "Finite-time formation control for first-order multi-agent systems with region constraints," *Frontiers of Information Technology & Electronic Engineering*, vol. 22, no. 1, pp. 134–140, 2021.
- [8] J. B. Flores-Resendiz, D. Avilés, and E. Aranda-Bricaire, "Formation control for second-order multi-agent systems with collision avoidance," *Machines*, vol. 11, no. 2, p. 208, 2023.
- [9] E. N. no, A. Loría, T. Hernández, M. Maghenem, and E. Panteley, "Consensus-based formation control of nonholonomic robots without velocity measurements," in *Proceedings of the 2020 IEEE Conference on Decision and Control*, 2020, pp. 823–828.
- [10] E. N. no, A. Loría, A. I. Paredes, and T. Hernández, "Consensus-based formation control of multiple nonholonomic vehicles under input constraints," *IEEE Control Systems Letters*, vol. 6, pp. 2767–2772, 2022.
- [11] A. Loría, E. N. no, and E. Panteley, "Observerless output-feedback consensus-based formation control of second-order nonholonomic systems," *IEEE Transactions on Automatic Control*, vol. 67, no. 10, pp. 5587–5594, 2021.
- [12] A. M. Bloch, *Nonholonomic Mechanics and Control*. Springer, 2003.
- [13] J. P. Desai, J. P. Ostrowski, and V. Kumar, "Modeling and control of formations of nonholonomic mobile robots," *IEEE Transactions on Robotics and Automation*, vol. 17, no. 6, pp. 905–908, 2001.
- [14] Z. Li, Y. Wang, and Q. Zhang, "Artificial potential-based formation control with collision and obstacle avoidance for second-order multi-agent systems," in *Proceedings of the IEEE Conference on Decision and Control*, 2021.
- [15] Z. Zhao, Y. Liu, and H. Wang, "An improved artificial potential field method for formation control and obstacle avoidance of the multi-agents systems," in *Proceedings of the IEEE International Conference*, 2024.
- [16] Z. Chen, Y. Li, and X. Wang, "Coordinated obstacle avoidance of multi-robot based on improved artificial potential field method and consistency protocol," *Journal of Marine Science and Engineering*, vol. 11, no. 6, p. 1157, 2023.
- [17] A. Kouloug and A. Pakniyat, "Distributed adaptive consensus with obstacle and collision avoidance for networks of heterogeneous multi-agent systems," *arXiv preprint arXiv:2410.08440*, 2024.